

INTERNATIONAL SCHOOL

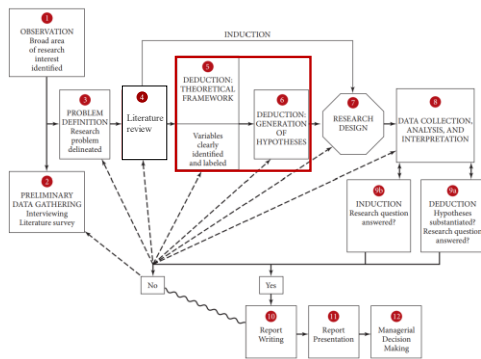
A theoretical framework Hypothesis Development

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RESEARCH WORKFLOW



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LEARNING OBJECTIVES

- 1) Understand the variable
- 2) Understand types of variable
- 3) Understand the theoretical framework
- 4) Understand how to develop a theoretical framework

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Definition of variable?



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VARIABLES

- A variable is an unknown quality that the researcher wants to measure in the research project.
- The variable must be measurable.
- The variable can vary from one object to another, or change over time.

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VARIABLES

- Examples of variables: "The Impact of coffee consumption on VNUIS student concentration in 2026"

Variable Name	Specific Value / Measurement Scale
Coffee Intake	0ml (Control group), 100ml, 200ml
Concentration Level	Scale from 1 (Very distracted) to 10 (Highly focused)
Time of Consumption	Morning, Afternoon, Evening
Blood Caffeine Concentration	Low, Medium, High
Previous Night's Sleep	Number of sleep hours (e.g., fixed at 7 hours)
Type of Coffee	Black coffee, Milk coffee, Espresso

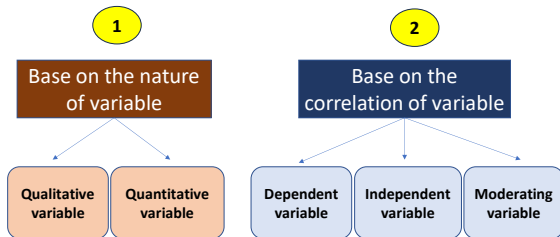
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TYPE OF VARIABLES

- Two main ways to divide the variable

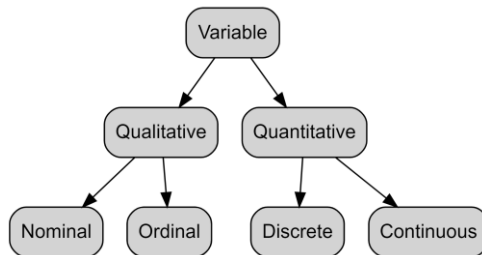


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Base on the nature of variable



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Qualitative variable

- Qualitative** variables (also referred as categorical variables) are variables that are **not numerical**.
- A **qualitative** variable is a variable that takes as its values modalities, **categories** or even levels,
- Example:**
Gender: Male or female
Marital status: Married, single, divorced, widowed
Educational level: High school, college, graduate school
Type of employment: Full-time, part-time, self-employed, unemployed

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Qualitative variable

NOMINAL

- A **qualitative nominal** variable is a qualitative variable where **no ordering** is possible or implied in the levels.
- 2 type of nominal:
 - Two levels (**Binary variable**): Smoke? - **Yes/No**
 - A large number of levels: Type of employment - **Full-time, part-time, self-employed, unemployed**

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Qualitative variable

ORDINAL

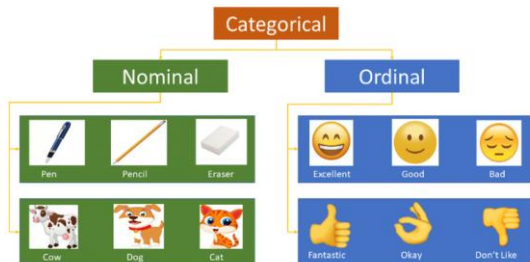
- A **qualitative ordinal** variable is a qualitative variable with an **order implied in the levels**.
- Need to arrange from smallest to largest, or vice versa.
- **Example: Highest education level**
 1. Primary school and below
 2. Secondary school
 3. High school
 4. University and above

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Qualitative variable



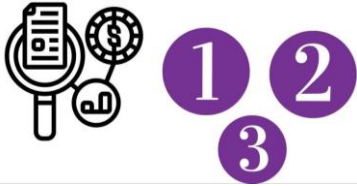
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Quantitative variable

- A **quantitative** variable is a variable that reflects a notion of **magnitude**, that is, if the values it can take are **numbers**. A quantitative variable represents thus a measure and is numerical.



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Quantitative variable

DISCRETE

- Quantitative discrete** variables are variables for which the values it can take are **countable** and have a **finite number of possibilities**. The values are often (but not always) integers.
- Example:**
 - Number of children per family (1, 2, 3, 4, ...)
 - Number of students in a class (35, 40, 42, ...)
 - Number of citizens of a city (100.000, 200.000, ...)

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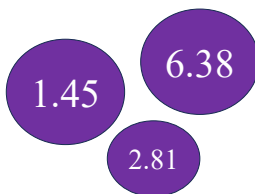
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Quantitative variable

CONTINUOUS

- Quantitative continuous** variables are variables for which the values are **not countable** and have an **infinite number of possibilities**.

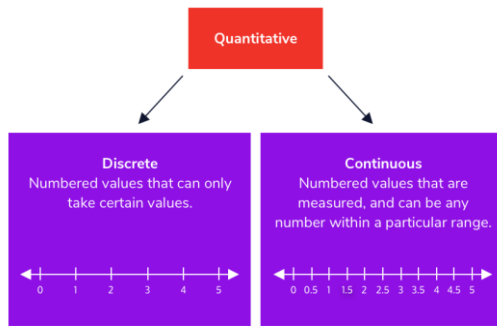


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Quatitative variable



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Exercise 1

Please categorize each variable in the following list as a **Qualitative variable** (nominal or ordinal) or **Quantitative variable** (discrete or continuous):

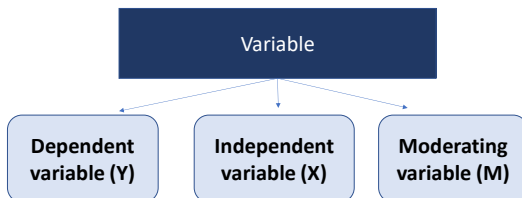
1. Age of participants
2. Time spent studying
3. Test scores
4. Gender
5. Career
6. Marital status
7. Experimental group vs. control group
8. Revenue
9. Customer Satisfaction
10. Supply Chain Efficiency

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Base on the correlation of variable



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Dependent variable

- The dependent variable (Y) is the variable of primary interest to the researcher
- ✓ Through the analysis of the dependent variable, it is possible to find answers or solutions to the problem
- ✓ Thus, the **researcher's goal** is to **understand and describe** the dependent variable

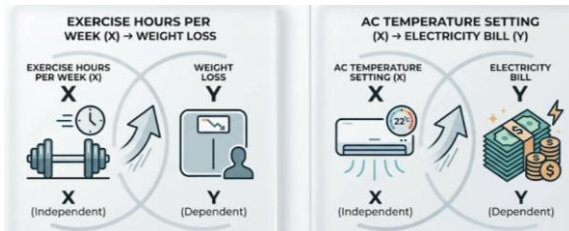
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Independent variable

- Independent variable (X) influences the dependent variable in either a positive or negative way
- When the independent variable is present, the dependent variable is also present:
 - ✓ (1) with each unit of increase in the independent variable, there is an increase or decrease in the dependent variable
 - ✓ (2) a change in the independent variable causes a change in the dependent variable

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Example



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Example



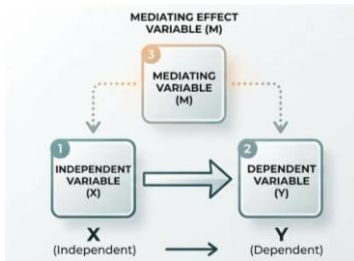
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3. Moderating variable

- A Moderating variable is a factor that changes the strength or direction of the relationship between an Independent variable and a Dependent variable.

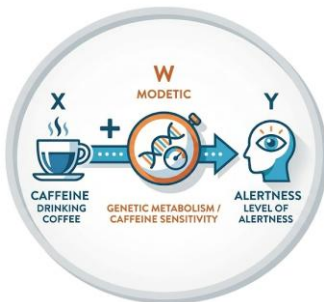


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Example



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Exercise 2

Please find out the Independent variable (X), the Dependent variable (Y), and the Moderating variable (M) in these cases:

1. The company assumes that a salary increase enhances employee motivation. However, while younger staff are strongly motivated by money, the effect of a raise is much weaker for employees near retirement age.
2. A study examines whether listening to music (X) improves student concentration (Y). The findings reveal that for students who prefer instrumental music, concentration increases significantly; however, for those who prefer vocal music, it acts as a distraction.

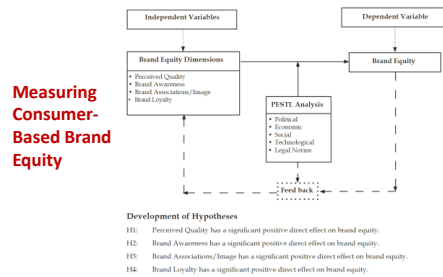
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THEORETICAL FRAMEWORK

- The theoretical framework is the foundation on which the research project is built



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The need for a theoretical framework

- Process of building a theoretical framework:

1) Definitions of the variables

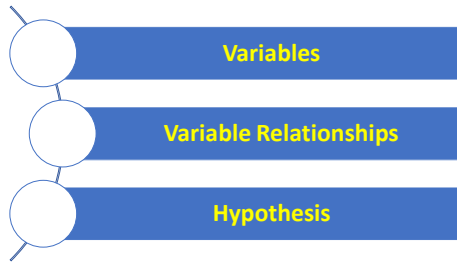
2) Create a theory that explains relationships between the variables in the diagram

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The components of the theoretical framework



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The components of the theoretical framework

▪ Variables:

- ✓ The variables considered relevant to the study should be **clearly identified and labeled** in the discussions

▪ Variable Relationships:

- ✓ The discussions should state how **two or more variables** are related to one another
- ✓ This should be done for the important relationships that are theorized to exist among the variables.

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The components of the theoretical framework

▪ Hypothesis:

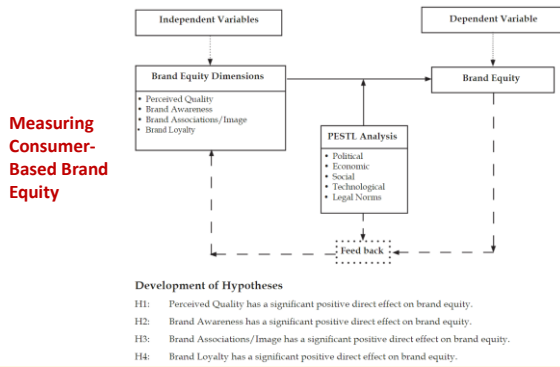
- ✓ A Hypothesis is the **expected relationship** between two or more variables.
- ✓ It is an 'educated guess' based on existing theory or observation.
- ✓ Directional vs. Non-directional Hypothesis

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Example (cont)



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Exercise 3

1. Choose any research topic for each group
2. Building a theoretical framework for that study

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